

The way meromorphic functions act on closed loops in the complex-projective space $\mathbb{CP}^1$

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Abstract

We study meromorphic functions on $\mathbb{CP}^1$ through the divisor structure determined by their zeros and poles. The main result shows that on $\mathbb{CP}^1$ the subgroup of principal divisors coincides with the kernel of the degree map. From this it follows that the divisor class group is isomorphic to $\mathbb{Z}$, and consequently so is the Picard group.

Contents

1	Short Introduction	2
2	Main Result	2
3	Algebraic and Analytic Preliminaries	2
3.1	The polynomial ring and the rational function field	2
3.2	Holomorphic and meromorphic functions	3
3.3	Zeros and local factorization of meromorphic functions	4
4	The Riemann Sphere and the Projective Line	5
4.1	The complex projective line	5
4.2	The standard affine charts	5
4.3	The point at infinity	6
4.4	Meromorphic functions on $\mathbb{CP}^1$	6
5	Orders and Divisors on $\mathbb{CP}^1$	7
5.1	Order at a finite point	7
5.2	The divisor group	7
5.3	The divisor of a rational function	8
5.4	Principal divisors	8
5.5	Basic examples	8
5.6	The degree of a divisor	9
6	The Exact Sequences	9
6.1	The divisor map revisited	9
6.2	The first exact sequence	10
6.3	The divisor class group	10
6.4	The second exact sequence	10
7	Principal Divisors and Degree on $\mathbb{CP}^1$	11
7.1	The inclusion $\text{im}(\text{Div}) \subseteq \ker(\text{deg})$	11
7.2	Every degree-zero divisor is principal	11
7.3	The main theorem	12
8	The Divisor Class Group of $\mathbb{CP}^1$	12
8.1	Degree on divisor classes	13
8.2	Surjectivity of the degree map	13
8.3	Injectivity of the degree map	13
8.4	The main theorem	13
8.5	Alternative proof via the First Isomorphism Theorem	14
8.6	Interpretation	14
9	From Divisor Classes to the Picard Group	14
9.1	The line bundle associated to a divisor	15
9.2	Principal divisors and the trivial bundle	15
9.3	From divisor classes to line bundles	15
9.4	Computation of the Picard group of $\mathbb{CP}^1$	16
10	Summary	16

1 Short Introduction

In mathematics, the classification of algebraic structures such as groups and rings often reveals hidden structure in the objects under study. In this paper, we apply this philosophy to meromorphic functions on the complex projective line $\mathbb{CP}^1$ by defining their divisor through the zero–pole configuration and then studying the quotient structures that arise from this construction.

2 Main Result

The aim is to classify functions through the divisor structure endowed by their zeros and poles, that is, the points at which the function vanishes or blows up to infinity. However, not all complex functions are polynomials. In fact, we can look at a larger class of functions, namely the rational functions, and more generally the meromorphic functions. To do this rigorously, we first need to understand some local notions about complex functions by studying their behaviour near zeros and poles, which will be defined in this paper. This requires a small introduction to Laurent series and the local factorization of meromorphic functions.

Furthermore, we extend the affine complex line $\mathbb{C}$ to the complex projective line $\mathbb{CP}^1$, which endows our space with a point at infinity. In this way, a polynomial is no longer seen only as a function with roots in $\mathbb{C}$, but also as a meromorphic function having a pole at infinity.

We develop this zero–pole configuration into a global algebraic object: the divisor of a meromorphic function. This leads naturally to the subgroup of principal divisors and to the divisor class group. The main structural theorem reached in this paper is that on $\mathbb{CP}^1$,

$$\text{Prin}(\mathbb{CP}^1) = \ker(\deg : \text{Div}(\mathbb{CP}^1) \rightarrow \mathbb{Z}),$$

from which it follows that

$$\text{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

A further consequence is the classical identification

$$\text{Pic}(\mathbb{CP}^1) \cong \mathbb{Z},$$

see [4, Chapter II, §6], [5, Tag 0BXJ].

Although these results on $\mathbb{CP}^1$ are simple and elegant, they hide a much deeper structure. They already suggest the role of Picard groups on general smooth curves, where divisor classes are no longer determined only by degree, and where principal divisors need not exhaust all divisors of degree zero.

What follows is therefore a construction of the divisor map, the group of principal divisors, the divisor class group, and the Picard group in this basic setting, together with the methods needed to obtain these results.

3 Algebraic and Analytic Preliminaries

We introduce the notions needed to study rational and meromorphic functions before passing to divisors. In particular, we want to understand how zeros and poles are measured locally, since these local integers will later become the coefficients of a divisor.

3.1 The polynomial ring and the rational function field

We introduce the polynomial ring and its field of fractions. The polynomial ring over the complex numbers is

$$\mathbb{C}[z] = \left\{ \sum_{j=0}^n a_j z^j : a_j \in \mathbb{C}, n \geq 0 \right\}.$$

The field of rational functions over the complex numbers is

$$\mathbb{C}(z) = \left\{ \frac{P(z)}{Q(z)} : P(z), Q(z) \in \mathbb{C}[z], Q(z) \neq 0 \right\}.$$

This set is a field because $\mathbb{C}[z]$ is an integral domain and $\mathbb{C}(z)$ is the field of fractions of $\mathbb{C}[z]$.

Proposition 3.1. Every nonzero rational function in $\mathbb{C}(z)$ may be written as

$$f(z) = c \frac{\prod_{i=1}^r (z - a_i)^{m_i}}{\prod_{j=1}^s (z - b_j)^{n_j}}$$

where $c \in \mathbb{C}^\times$, the points $a_i, b_j \in \mathbb{C}$ are pairwise distinct, and the integers m_i, n_j are positive.

Proof. Every nonzero element of $\mathbb{C}(z)$ is of the form $f = P/Q$ with $P, Q \in \mathbb{C}[z]$ and $Q \neq 0$. Since $\mathbb{C}$ is algebraically closed, the Fundamental Theorem of Algebra implies that both P and Q factor into linear factors:

$$P(z) = c \prod_{i=1}^r (z - a_i)^{m_i}, \quad Q(z) = \prod_{j=1}^s (z - b_j)^{n_j},$$

for suitable $c \in \mathbb{C}^\times$, distinct points $a_i, b_j \in \mathbb{C}$, and positive integers m_i, n_j . Therefore

$$f(z) = c \frac{\prod_{i=1}^r (z - a_i)^{m_i}}{\prod_{j=1}^s (z - b_j)^{n_j}}.$$

□

Given this factorization, it is natural to classify polynomials by their local behaviour around zeros and poles.

Definition 3.2. Let $P(z) \in \mathbb{C}[z]$ and let $a \in \mathbb{C}$. We say that a is a zero of multiplicity m of P if

$$P(z) = (z - a)^m Q(z),$$

where $Q(z) \in \mathbb{C}[z]$ and $Q(a) \neq 0$.

Proposition 3.3. If $P(z) = c \prod_{j=1}^r (z - a_j)^{m_j}$, then the degree of P is $\deg(P) = \sum_{j=1}^r m_j$.

Proof. The degree of a product of nonzero polynomials is the sum of the degrees. Since each factor $(z - a_j)^{m_j}$ has degree m_j and $c \in \mathbb{C}^\times$ has degree 0, we obtain

$$\deg(P) = \sum_{j=1}^r \deg((z - a_j)^{m_j}) = \sum_{j=1}^r m_j.$$

□

3.2 Holomorphic and meromorphic functions

We extend the class of functions from polynomials and rational functions to the analytic setting. This is convenient because meromorphic functions admit local expansions that encode poles and their order.

Definition 3.4. Let $U \subseteq \mathbb{C}$ be open. A function $f : U \rightarrow \mathbb{CP}^1$ is called meromorphic on U if it is holomorphic on U except at isolated points, where it has poles.

A fundamental theorem of complex analysis states the following [1, Chapter 5].

Theorem 3.5 (Laurent Expansion). *Let $A = \{z \in \mathbb{C} : 0 < |z - a| < R\}$ be a punctured disc, and let f be holomorphic on A . Then there exists a unique series $\sum_{n=-\infty}^{\infty} c_n(z - a)^n$ such that*

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z - a)^n$$

for all $z \in A$.

The Laurent expansion allows one to distinguish removable singularities, poles, and essential singularities according to the negative-power part of the series.

Definition 3.6. Let f be meromorphic in a punctured neighbourhood of $a \in \mathbb{C}$. We say that a is a pole of order m if the Laurent expansion of f around a has the form

$$f(z) = \sum_{n=-m}^{\infty} c_n(z - a)^n, \quad c_{-m} \neq 0.$$

Proposition 3.7. If a is a pole of order m of f , then there exists a holomorphic function u near a such that

$$f(z) = (z - a)^{-m}u(z), \quad u(a) \neq 0.$$

Proof. Since a is a pole of order m , the Laurent expansion of f around a has the form

$$f(z) = \sum_{n=-m}^{\infty} c_n(z - a)^n, \quad c_{-m} \neq 0.$$

Factor out $(z - a)^{-m}$:

$$f(z) = (z - a)^{-m} \sum_{n=-m}^{\infty} c_n(z - a)^{n+m}.$$

Define

$$u(z) := \sum_{n=-m}^{\infty} c_n(z - a)^{n+m}.$$

Then u is holomorphic near a and $u(a) = c_{-m} \neq 0$. Therefore

$$f(z) = (z - a)^{-m}u(z), \quad u(a) \neq 0.$$

□

We are now ready to set up our divisor classification for arbitrary meromorphic functions.

3.3 Zeros and local factorization of meromorphic functions

We give the local form of a meromorphic function near a zero or pole,

Proposition 3.8. Let f be holomorphic in a neighbourhood of $a \in \mathbb{C}$ and suppose that f is not identically zero. Then either $f(a) \neq 0$, or there exists a unique integer $m \geq 1$ and a holomorphic function u such that

$$f(z) = (z - a)^m u(z), \quad u(a) \neq 0.$$

Proof. Since f is holomorphic near a , it has a Taylor expansion

$$f(z) = \sum_{n=0}^{\infty} c_n(z - a)^n.$$

If $f(a) \neq 0$, then there is nothing to prove. Otherwise, since f is not identically zero, there exists a smallest integer $m \geq 1$ such that $c_m \neq 0$. Then

$$f(z) = \sum_{n=m}^{\infty} c_n(z-a)^n = (z-a)^m \sum_{n=m}^{\infty} c_n(z-a)^{n-m}.$$

Define

$$u(z) := \sum_{n=m}^{\infty} c_n(z-a)^{n-m}.$$

Then u is holomorphic near a and $u(a) = c_m \neq 0$, hence $f(z) = (z-a)^m u(z)$, $u(a) \neq 0$.

To prove uniqueness, suppose also $f(z) = (z-a)^n v(z)$, $v(a) \neq 0$. Assume without loss of generality that $m < n$. Then $u(z) = (z-a)^{n-m} v(z)$, which implies $u(a) = 0$, a contradiction. Therefore $m = n$, and then $u = v$. $\square$

Definition 3.9. If $f(z) = (z-a)^m u(z)$, $u(a) \neq 0$, then a is called a zero of order m of f .

Remark 3.10. Together with Proposition 3.7, this shows that zeros and poles are both measured locally by integers. This is the key fact that makes the divisor construction possible in the next sections.

Next, we construct the Riemann sphere with the goal of including the point at infinity, since polynomials can then be regarded as having poles at infinity, and rational functions may also have zeros at infinity.

4 The Riemann Sphere and the Projective Line

For the arguments of this paper, we do not need a broad development of affine geometry. Instead, we focus only on the construction of the complex projective line, its topological and analytic structure, and its identification with the Riemann sphere.

4.1 The complex projective line

Definition 4.1. The complex projective line is defined as $\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{(0,0)\}) / \sim$ where the equivalence relation is given by $(X, Y) \sim (\lambda X, \lambda Y)$ for every $\lambda \in \mathbb{C}^\times$. An equivalence class is denoted by $[X : Y]$.

Since $\mathbb{CP}^1$ is defined as a quotient, it carries the quotient topology induced by the canonical projection

$$\pi : \mathbb{C}^2 \setminus \{(0,0)\} \rightarrow \mathbb{CP}^1, \quad (X, Y) \mapsto [X : Y].$$

A subset $U \subseteq \mathbb{CP}^1$ is said to be open if and only if $\pi^{-1}(U)$ is open in $\mathbb{C}^2 \setminus \{(0,0)\}$. This gives $\mathbb{CP}^1$ a natural topological structure.

4.2 The standard affine charts

To describe $\mathbb{CP}^1$ concretely, we use two standard open subsets:

$$U_0 = \{[X : Y] \in \mathbb{CP}^1 : Y \neq 0\}, \quad U_1 = \{[X : Y] \in \mathbb{CP}^1 : X \neq 0\}.$$

On U_0 , define

$$\phi_0 : U_0 \rightarrow \mathbb{C}, \quad [X : Y] \mapsto \frac{X}{Y}.$$

On U_1 , define

$$\phi_1 : U_1 \rightarrow \mathbb{C}, \quad [X : Y] \mapsto \frac{Y}{X}.$$

Proposition 4.2. The maps ϕ_0 and ϕ_1 are bijections. Hence each of the sets U_0 and U_1 is identified with a copy of $\mathbb{C}$.

Proof. We prove the statement for ϕ_0 ; the proof for ϕ_1 is analogous. Define

$$\psi_0 : \mathbb{C} \rightarrow U_0, \quad z \mapsto [z : 1].$$

Then $\phi_0(\psi_0(z)) = \phi_0([z : 1]) = \frac{z}{1} = z$, so $\phi_0 \circ \psi_0 = \text{id}_{\mathbb{C}}$. Conversely, if $[X : Y] \in U_0$, then $\psi_0(\phi_0([X : Y])) = \psi_0(\frac{X}{Y}) = [\frac{X}{Y} : 1] = [X : Y]$. Therefore $\psi_0 \circ \phi_0 = \text{id}_{U_0}$, so ϕ_0 is bijective. Similarly, ϕ_1 is bijective with inverse $\psi_1(z) = [1 : z]$. $\square$

The transition map on the overlap $U_0 \cap U_1$ is given by

$$\phi_1 \circ \phi_0^{-1}(z) = \frac{1}{z}, \quad z \in \mathbb{C}^\times.$$

Since this transition map is holomorphic, the two charts define a complex manifold structure on $\mathbb{CP}^1$.

4.3 The point at infinity

Using the chart U_0 , every point of U_0 may be written uniquely as $[z : 1]$ with $z \in \mathbb{C}$. Thus U_0 is identified with $\mathbb{C}$. There is one additional point not contained in this chart, namely $[1 : 0]$. This point is called the point at infinity and is denoted by ∞ . Therefore, $\mathbb{CP}^1 = \mathbb{C} \cup \{\infty\}$.

This is the usual Riemann sphere. The chart U_1 provides a local coordinate near ∞ , namely $w = \frac{1}{z}$, which is essential for defining poles and orders at infinity.

4.4 Meromorphic functions on $\mathbb{CP}^1$

Now that $\mathbb{CP}^1$ has been given a complex manifold structure, it makes sense to define holomorphic and meromorphic functions on it.

Definition 4.3. A function

$$f : \mathbb{CP}^1 \rightarrow \mathbb{CP}^1$$

is called meromorphic if, in each local chart, it is represented by a meromorphic function of one complex variable.

In particular, on the affine chart $U_0 \cong \mathbb{C}$, such a function is represented by a meromorphic function in the usual sense. Near the point ∞ , it is represented in the local coordinate $w = 1/z$.

Theorem 4.4. Every meromorphic function on $\mathbb{CP}^1$ is rational. More precisely, if

$$f : \mathbb{CP}^1 \rightarrow \mathbb{CP}^1$$

is meromorphic, then on the affine chart $\mathbb{C} \subseteq \mathbb{CP}^1$ it is given by an element of $\mathbb{C}(z)$.

Proof. This is a standard result in the theory of compact Riemann surfaces; see [2, Chapter 7] or [3, Chapter III]. Briefly, the restriction of f to the affine chart $\mathbb{C}$ has only finitely many poles, because poles are isolated and $\mathbb{CP}^1$ is compact. After multiplying by a polynomial that cancels these poles, one obtains an entire function of polynomial growth, hence a polynomial. Therefore f is a quotient of two polynomials, i.e. an element of $\mathbb{C}(z)$. $\square$

Notice how a polynomial $P(z) \in \mathbb{C}[z]$ defines a holomorphic function on $\mathbb{C}$, but when viewed on $\mathbb{CP}^1$ it becomes a meromorphic function with a pole at infinity. Thus $\mathbb{CP}^1$ gives us a natural space where zeros and poles may be studied together in one global geometric framework.

5 Orders and Divisors on $\mathbb{CP}^1$

We now pass from the local analytic behaviour of meromorphic functions to a global algebraic object. By Theorem 4.4, meromorphic functions on $\mathbb{CP}^1$ may be identified with rational functions, and near a zero or a pole the function is locally controlled by an integer. We now assemble these local integers into a formal sum, called the divisor of a function.

5.1 Order at a finite point

Definition 5.1. Let $f \in \mathbb{C}(z)^\times$ and let $a \in \mathbb{C}$. We define the order of f at a , denoted by $\text{ord}_a(f)$, as follows:

- if f has a zero of order m at a , then $\text{ord}_a(f) = m$;
- if f has a pole of order m at a , then $\text{ord}_a(f) = -m$;
- if f has neither a zero nor a pole at a , then $\text{ord}_a(f) = 0$.

Equivalently, if near a one has a local factorization

$$f(z) = (z - a)^m u(z), \quad u(a) \neq 0,$$

then $\text{ord}_a(f) = m$. The positive order corresponds to a zero, negative order corresponds to a pole, and zero order means that the function is finite and nonzero at the point. Since we are working on $\mathbb{CP}^1 = \mathbb{C} \cup \{\infty\}$, we must also define the order at the point at infinity.

Definition 5.2. Let $f \in \mathbb{C}(z)^\times$. The order of f at infinity is defined by

$$\text{ord}_\infty(f) = \text{ord}_0(f(1/w)),$$

where the right-hand side is computed in the local coordinate $w = 1/z$.

Example 5.3. For the function $f(z) = z$, we have $f(1/w) = 1/w$, so $\text{ord}_\infty(z) = -1$. Thus z has a simple pole at infinity.

Example 5.4. For the function $f(z) = z^n$, we have $f(1/w) = w^{-n}$, hence $\text{ord}_\infty(z^n) = -n$.

5.2 The divisor group

Definition 5.5. The divisor group on $\mathbb{CP}^1$ is

$$\text{Div}(\mathbb{CP}^1) := \bigoplus_{p \in \mathbb{CP}^1} \mathbb{Z} \cdot p.$$

An element of $\text{Div}(\mathbb{CP}^1)$ is therefore a finite formal sum

$$D = \sum_{p \in \mathbb{CP}^1} n_p p, \quad n_p \in \mathbb{Z},$$

where all but finitely many of the coefficients n_p are zero.

Remark 5.6. The group $\text{Div}(\mathbb{CP}^1)$ is abelian under coefficientwise addition:

$$\left(\sum_{p \in \mathbb{CP}^1} n_p p \right) + \left(\sum_{p \in \mathbb{CP}^1} m_p p \right) = \sum_{p \in \mathbb{CP}^1} (n_p + m_p) p.$$

5.3 The divisor of a rational function

Definition 5.7. Let $f \in \mathbb{C}(z)^\times$. The divisor of f is defined by

$$\operatorname{div}(f) := \sum_{p \in \mathbb{CP}^1} \operatorname{ord}_p(f) p.$$

Since a rational function has only finitely many zeros and poles, this sum is finite and therefore defines an element of $\operatorname{Div}(\mathbb{CP}^1)$.

5.4 Principal divisors

Definition 5.8. A divisor on $\mathbb{CP}^1$ is called principal if it is the divisor of a nonzero rational function. The subgroup of the divisor group containing all principal divisors is denoted by

$$\operatorname{Prin}(\mathbb{CP}^1) = \operatorname{im}(\operatorname{div} : \mathbb{C}(z)^\times \rightarrow \operatorname{Div}(\mathbb{CP}^1)).$$

Thus principal divisors are exactly those divisors which arise from meromorphic functions on $\mathbb{CP}^1$.

Proposition 5.9. The map $\operatorname{Div} : \mathbb{C}(z)^\times \rightarrow \operatorname{Div}(\mathbb{CP}^1)$ is a group homomorphism.

Proof. Let $f, g \in \mathbb{C}(z)^\times$ and let $p \in \mathbb{CP}^1$. Near p , write $f = (t_p)^m u$, $g = (t_p)^n v$, where t_p is a local parameter at p , u and v are holomorphic and nonvanishing near p , and $m = \operatorname{ord}_p(f)$, $n = \operatorname{ord}_p(g)$. Then $fg = (t_p)^{m+n}(uv)$, and since uv is holomorphic and nonvanishing near p , it follows that $\operatorname{ord}_p(fg) = \operatorname{ord}_p(f) + \operatorname{ord}_p(g)$. Therefore

$$\operatorname{Div}(fg) = \sum_{p \in \mathbb{CP}^1} \operatorname{ord}_p(fg) p = \sum_{p \in \mathbb{CP}^1} (\operatorname{ord}_p(f) + \operatorname{ord}_p(g)) p,$$

hence

$$\operatorname{Div}(fg) = \operatorname{Div}(f) + \operatorname{Div}(g).$$

□

5.5 Basic examples

Proposition 5.10. For every $a \in \mathbb{C}$, $\operatorname{Div}(z - a) = (a) - (\infty)$.

Proof. The function $z - a$ has a simple zero at a and no other finite zeros or poles. Near infinity, using the coordinate $w = 1/z$, one has

$$z - a = \frac{1}{w} - a = \frac{1 - aw}{w},$$

and since $1 - aw$ is holomorphic and nonzero at $w = 0$, it follows that $z - a$ has a simple pole at infinity. Therefore

$$\operatorname{Div}(z - a) = (a) - (\infty).$$

□

Proposition 5.11. For every integer $n \geq 1$, $\operatorname{Div}(z^n) = n(0) - n(\infty)$.

Proof. The function z^n has a zero of order n at 0 and no other finite zeros or poles. Near infinity, using $w = 1/z$, one has $z^n = w^{-n}$, so z^n has a pole of order n at ∞ . Therefore

$$\operatorname{Div}(z^n) = n(0) - n(\infty).$$

□

Proposition 5.12. Let

$$P(z) = c \prod_{j=1}^r (z - a_j)^{m_j}$$

be a nonzero polynomial. Then

$$\operatorname{Div}(P) = \sum_{j=1}^r m_j(a_j) - (\deg P)(\infty).$$

Proof. By Proposition 5.9, Proposition 5.10, and the fact that $\operatorname{Div}(c) = 0$ for every $c \in \mathbb{C}^\times$, we obtain

$$\operatorname{Div}(P) = \sum_{j=1}^r \operatorname{Div}((z - a_j)^{m_j}) = \sum_{j=1}^r (m_j(a_j) - m_j(\infty)).$$

Hence

$$\operatorname{Div}(P) = \sum_{j=1}^r m_j(a_j) - \left(\sum_{j=1}^r m_j \right) (\infty).$$

Using Proposition 3.3, this becomes

$$\operatorname{Div}(P) = \sum_{j=1}^r m_j(a_j) - (\deg P)(\infty).$$

□

5.6 The degree of a divisor

Definition 5.13. The degree map is the homomorphism $\deg : \operatorname{Div}(\mathbb{CP}^1) \rightarrow \mathbb{Z}$ defined by

$$\deg \left(\sum_{p \in \mathbb{CP}^1} n_p p \right) = \sum_{p \in \mathbb{CP}^1} n_p.$$

Example 5.14. If $D = 2(a) - 3(b) + (\infty)$, then $\deg(D) = 2 - 3 + 1 = 0$.

6 The Exact Sequences

At this stage, two natural questions arise:

- Which divisors actually come from meromorphic functions?
- How far is an arbitrary divisor from being of that form?

These two questions lead naturally to two exact sequences.

6.1 The divisor map revisited

Recall that every nonzero rational function on $\mathbb{CP}^1$ defines a divisor, and hence we have a map $\operatorname{Div} : \mathbb{C}(z)^\times \rightarrow \operatorname{Div}(\mathbb{CP}^1)$. By Proposition 5.9, this map is a group homomorphism. Its image is, by definition, the subgroup of principal divisors: $\operatorname{Prin}(\mathbb{CP}^1) = \operatorname{im}(\operatorname{Div})$.

Proposition 6.1. The kernel of the divisor map is the multiplicative group of nonzero constants:

$$\ker(\operatorname{Div}) = \mathbb{C}^\times.$$

Proof. If $c \in \mathbb{C}^\times$, then c has no zeros and no poles on $\mathbb{CP}^1$, so $\text{Div}(c) = 0$. Thus $\mathbb{C}^\times \subseteq \ker(\text{Div})$.

Conversely, let $f \in \mathbb{C}(z)^\times$ satisfy $\text{Div}(f) = 0$. Then f has no zeros and no poles on $\mathbb{CP}^1$. In particular, f is holomorphic everywhere on $\mathbb{CP}^1$. Since $\mathbb{CP}^1$ is compact, every holomorphic function on $\mathbb{CP}^1$ is constant; see [2, Chapter 7]. Therefore $f \in \mathbb{C}^\times$. Hence

$$\ker(\text{Div}) = \mathbb{C}^\times.$$

□

This means that two rational functions have the same divisor if and only if they differ by multiplication by a nonzero constant. Indeed, if $\text{Div}(f) = \text{Div}(g)$, then $\text{Div}\left(\frac{f}{g}\right) = 0$, and therefore f/g is constant by Proposition 6.1.

6.2 The first exact sequence

The previous discussion yields the first exact sequence:

Proposition 6.2. The sequence

$$1 \rightarrow \mathbb{C}^\times \rightarrow \mathbb{C}(z)^\times \xrightarrow{\text{Div}} \text{Prin}(\mathbb{CP}^1) \rightarrow 1$$

is exact.

Proof. The map $\mathbb{C}^\times \rightarrow \mathbb{C}(z)^\times$ is the natural inclusion, so it is injective. By Proposition 6.1, $\ker(\text{Div}) = \mathbb{C}^\times$, so the sequence is exact at $\mathbb{C}(z)^\times$. Finally, the map $\text{Div} : \mathbb{C}(z)^\times \rightarrow \text{Prin}(\mathbb{CP}^1)$ is surjective by Definition 5.8. Hence the sequence is exact at $\text{Prin}(\mathbb{CP}^1)$. □

Remark 6.3. This exact sequence expresses the fact that principal divisors are exactly rational functions modulo nonzero constants: $\text{Prin}(\mathbb{CP}^1) \cong \frac{\mathbb{C}(z)^\times}{\mathbb{C}^\times}$. This is just the First Isomorphism Theorem.

6.3 The divisor class group

Definition 6.4. The divisor class group of $\mathbb{CP}^1$ is defined by

$$\text{Cl}(\mathbb{CP}^1) = \frac{\text{Div}(\mathbb{CP}^1)}{\text{Prin}(\mathbb{CP}^1)}.$$

Two divisors D and E define the same class if and only if $D - E \in \text{Prin}(\mathbb{CP}^1)$.

Definition 6.5. Two divisors D and E are called linearly equivalent if $D - E \in \text{Prin}(\mathbb{CP}^1)$. In this case one writes $D \sim E$.

The proof of linear equivalence being an equivalence relation is left to the reader as an exercise.

6.4 The second exact sequence

Since principal divisors form a subgroup of the divisor group, this gives the second exact sequence:

Proposition 6.6. The sequence

$$0 \rightarrow \text{Prin}(\mathbb{CP}^1) \rightarrow \text{Div}(\mathbb{CP}^1) \rightarrow \text{Cl}(\mathbb{CP}^1) \rightarrow 0$$

is exact.

Proof. The first map is the natural inclusion of $\text{Prin}(\mathbb{CP}^1)$ into $\text{Div}(\mathbb{CP}^1)$, so it is injective. The second map is the quotient map

$$q : \text{Div}(\mathbb{CP}^1) \rightarrow \text{Cl}(\mathbb{CP}^1), \quad D \mapsto [D].$$

By definition of quotient group, $\ker(q) = \text{Prin}(\mathbb{CP}^1)$. Thus the sequence is exact at $\text{Div}(\mathbb{CP}^1)$. Finally, q is surjective by definition, since every divisor class has the form $[D]$ for some $D \in \text{Div}(\mathbb{CP}^1)$. $\square$

7 Principal Divisors and Degree on $\mathbb{CP}^1$

We now prove the key structural result

$$\text{Prin}(\mathbb{CP}^1) = \ker(\deg).$$

In other words, a divisor on $\mathbb{CP}^1$ is principal if and only if it has degree zero.

7.1 The inclusion $\text{im}(\text{Div}) \subseteq \ker(\deg)$

Proposition 7.1. For every nonzero rational function $f \in \mathbb{C}(z)^\times$, one has

$$\deg(\text{Div}(f)) = 0.$$

Equivalently,

$$\text{im}(\text{Div}) \subseteq \ker(\deg).$$

Proof. Let

$$f(z) = c \frac{\prod_{i=1}^r (z - a_i)^{m_i}}{\prod_{j=1}^s (z - b_j)^{n_j}}, \quad c \in \mathbb{C}^\times.$$

Then f has zeros of orders m_i at the points a_i , poles of orders n_j at the points b_j , and

$$\text{ord}_\infty(f) = \sum_{j=1}^s n_j - \sum_{i=1}^r m_i.$$

Therefore

$$\deg(\text{Div}(f)) = \sum_{i=1}^r m_i - \sum_{j=1}^s n_j + \text{ord}_\infty(f) = 0.$$

Thus every principal divisor has degree zero. $\square$

7.2 Every degree-zero divisor is principal

Proposition 7.2. Let

$$D = \sum_{k=1}^r n_k(a_k)$$

be a divisor on $\mathbb{CP}^1$, where $a_1, \dots, a_r \in \mathbb{C}$ are distinct points and

$$\deg(D) = \sum_{k=1}^r n_k = 0.$$

Then D is principal.

Proof. Consider the rational function

$$f(z) = \prod_{k=1}^r (z - a_k)^{n_k},$$

where negative exponents are interpreted as factors in the denominator.

We obtain:

$$\operatorname{Div}((z - a_k)^{n_k}) = n_k(a_k) - n_k(\infty).$$

Since the divisor map is a homomorphism,

$$\operatorname{Div}(f) = \sum_{k=1}^r (n_k(a_k) - n_k(\infty)) = \sum_{k=1}^r n_k(a_k) - \left(\sum_{k=1}^r n_k \right) (\infty).$$

Because $\deg(D) = \sum_{k=1}^r n_k = 0$, this simplifies to

$$\operatorname{Div}(f) = \sum_{k=1}^r n_k(a_k) = D.$$

Thus D is principal. □

7.3 The main theorem

Theorem 7.3. *On $\mathbb{CP}^1$ one has*

$$\operatorname{Prin}(\mathbb{CP}^1) = \ker(\deg).$$

Proof. By Proposition 7.1, every principal divisor has degree zero, so

$$\operatorname{Prin}(\mathbb{CP}^1) \subseteq \ker(\deg).$$

Conversely, let $D \in \ker(\deg)$. Then $\deg(D) = 0$, so by Proposition 7.2 the divisor D is principal. Hence

$$\ker(\deg) \subseteq \operatorname{Prin}(\mathbb{CP}^1).$$

Therefore

$$\operatorname{Prin}(\mathbb{CP}^1) = \ker(\deg).$$

□

This theorem shows that on the Riemann sphere the only obstruction to a divisor being principal is its degree. If the total signed multiplicity of the divisor is zero, then that divisor comes from an actual rational function. This is precisely what makes $\mathbb{CP}^1$ so special: on a general smooth curve, a divisor of degree zero need not be principal.

8 The Divisor Class Group of $\mathbb{CP}^1$

We then compute the divisor class group of $\mathbb{CP}^1$. By Theorem 7.3, we already know that $\operatorname{Prin}(\mathbb{CP}^1) = \ker(\deg)$. This allows us to describe the quotient

$$\operatorname{Cl}(\mathbb{CP}^1) = \frac{\operatorname{Div}(\mathbb{CP}^1)}{\operatorname{Prin}(\mathbb{CP}^1)}$$

explicitly in terms of the degree map.

8.1 Degree on divisor classes

Recall that the degree map is

$$\deg : \text{Div}(\mathbb{CP}^1) \rightarrow \mathbb{Z}, \quad \deg \left(\sum_{p \in \mathbb{CP}^1} n_p p \right) = \sum_{p \in \mathbb{CP}^1} n_p.$$

Proposition 8.1. The degree map descends to a well-defined group homomorphism

$$\overline{\deg} : \text{Cl}(\mathbb{CP}^1) \rightarrow \mathbb{Z}, \quad [D] \mapsto \deg(D).$$

Proof. Suppose that two divisors D and E represent the same class in $\text{Cl}(\mathbb{CP}^1)$. Then

$$[D] = [E] \iff D - E \in \text{Prin}(\mathbb{CP}^1).$$

By Theorem 7.3, $\text{Prin}(\mathbb{CP}^1) = \ker(\deg)$. Therefore $\deg(D - E) = 0$, and hence $\deg(D) = \deg(E)$. So the value of $\deg(D)$ depends only on the class $[D]$, and the map $\overline{\deg}([D]) = \deg(D)$ is well-defined. It is a group homomorphism because degree itself is a group homomorphism on $\text{Div}(\mathbb{CP}^1)$. $\square$

8.2 Surjectivity of the degree map

Proposition 8.2. The map

$$\overline{\deg} : \text{Cl}(\mathbb{CP}^1) \rightarrow \mathbb{Z}$$

is surjective.

Proof. Let $n \in \mathbb{Z}$. Choose any point $p \in \mathbb{CP}^1$. Then the divisor $n(p)$ has degree $\deg(n(p)) = n$, hence $\overline{\deg}([n(p)]) = n$. Since n was arbitrary, the map is surjective. $\square$

8.3 Injectivity of the degree map

Proposition 8.3. The map

$$\overline{\deg} : \text{Cl}(\mathbb{CP}^1) \rightarrow \mathbb{Z}$$

is injective.

Proof. Suppose that $\overline{\deg}([D]) = 0$. By definition of $\overline{\deg}$, this means $\deg(D) = 0$. Hence $D \in \ker(\deg)$. By Theorem 7.3, $\ker(\deg) = \text{Prin}(\mathbb{CP}^1)$. Therefore D is principal, and so its class in the quotient is trivial: $[D] = 0$ in $\text{Cl}(\mathbb{CP}^1)$. Thus the kernel of $\overline{\deg}$ is trivial. $\square$

8.4 The main theorem

Theorem 8.4. The divisor class group of the complex projective line is isomorphic to the additive group of integers:

$$\text{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

Proof. By Proposition 8.1, Proposition 8.2, and Proposition 8.3, the map

$$\overline{\deg} : \text{Cl}(\mathbb{CP}^1) \rightarrow \mathbb{Z}$$

is a well-defined bijective group homomorphism. Therefore it is an isomorphism. $\square$

8.5 Alternative proof via the First Isomorphism Theorem

Proposition 8.5. There is an isomorphism

$$\mathrm{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

Proof. Since

$$\mathrm{Prin}(\mathbb{CP}^1) = \ker(\deg),$$

we have

$$\mathrm{Cl}(\mathbb{CP}^1) = \frac{\mathrm{Div}(\mathbb{CP}^1)}{\mathrm{Prin}(\mathbb{CP}^1)} = \frac{\mathrm{Div}(\mathbb{CP}^1)}{\ker(\deg)}.$$

By the First Isomorphism Theorem,

$$\frac{\mathrm{Div}(\mathbb{CP}^1)}{\ker(\deg)} \cong \mathrm{im}(\deg).$$

For any integer n and any point $p \in \mathbb{CP}^1$ one has $\deg(n(p)) = n$, so the map $\deg$ is surjective and $\mathrm{im}(\deg) = \mathbb{Z}$. Therefore

$$\mathrm{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

□

8.6 Interpretation

This theorem shows that on $\mathbb{CP}^1$, divisor classes are completely classified by degree. Once one passes to divisor classes, the remaining information is encoded by a single integer.

Equivalently, every divisor on $\mathbb{CP}^1$ is linearly equivalent to one of the form $n(\infty)$ for a unique integer n .

Proposition 8.6. Every divisor $D \in \mathrm{Div}(\mathbb{CP}^1)$ is linearly equivalent to

$$(\deg D)(\infty).$$

Proof. Consider the divisor

$$D - (\deg D)(\infty).$$

Its degree is

$$\deg(D) - \deg(D) = 0.$$

Hence

$$D - (\deg D)(\infty) \in \ker(\deg) = \mathrm{Prin}(\mathbb{CP}^1).$$

Therefore

$$D \sim (\deg D)(\infty).$$

□

9 From Divisor Classes to the Picard Group

On a smooth curve, divisors do not only encode zeros and poles of meromorphic functions; they also determine line bundles. This leads to the Picard group, which is the natural geometric counterpart of the divisor class group.

Definition 9.1. A *line bundle* on a complex manifold X is a holomorphic vector bundle of rank 1 over X .

Definition 9.2. The *Picard group* of X , denoted by

$$\mathrm{Pic}(X),$$

is the group of isomorphism classes of holomorphic line bundles on X , with group operation given by tensor product.

9.1 The line bundle associated to a divisor

Let $D = \sum n_p p$ be a divisor on $\mathbb{CP}^1$. One may associate to D a sheaf, denoted by

$$\mathcal{O}(D),$$

whose local sections are meromorphic functions whose poles and zeros are controlled by the divisor D .

Definition 9.3. Let D be a divisor on $\mathbb{CP}^1$. The sheaf $\mathcal{O}(D)$ is defined by

$$\mathcal{O}(D)(U) = \{f \in \mathbb{C}(z) : \text{Div}(f)|_U + D|_U \geq 0\} \cup \{0\}$$

for each open set $U \subseteq \mathbb{CP}^1$.

Remark 9.4. For the purposes of this paper, the important point is not the full sheaf-theoretic definition, but the fact that each divisor D gives rise to a line bundle $\mathcal{O}(D)$.

9.2 Principal divisors and the trivial bundle

Proposition 9.5. If $D = \text{Div}(f)$ is a principal divisor, then

$$\mathcal{O}(D) \cong \mathcal{O},$$

where $\mathcal{O}$ denotes the trivial line bundle on $\mathbb{CP}^1$.

Proof. This is a standard result; see [4, Chapter II, Proposition 6.13]. The isomorphism is given locally by multiplication by the meromorphic function f^{-1} . $\square$

This means that divisors differing by a principal divisor define isomorphic line bundles. Therefore the assignment $D \mapsto \mathcal{O}(D)$ depends only on the divisor class of D .

9.3 From divisor classes to line bundles

Proposition 9.6. The map

$$D \mapsto \mathcal{O}(D)$$

induces a well-defined group homomorphism

$$\text{Cl}(\mathbb{CP}^1) \rightarrow \text{Pic}(\mathbb{CP}^1).$$

Proof. By Proposition 9.5, if $D - E$ is principal, then $\mathcal{O}(D - E) \cong \mathcal{O}$. Equivalently, $\mathcal{O}(D) \cong \mathcal{O}(E)$. Hence the construction depends only on the class of D in $\text{Cl}(\mathbb{CP}^1)$ and therefore induces a well-defined map $\text{Cl}(\mathbb{CP}^1) \rightarrow \text{Pic}(\mathbb{CP}^1)$. It is a group homomorphism because

$$\mathcal{O}(D + E) \cong \mathcal{O}(D) \otimes \mathcal{O}(E).$$

$\square$

On a smooth curve, this map is in fact an isomorphism.

Theorem 9.7. For a smooth complex curve X , there is a natural isomorphism

$$\text{Cl}(X) \cong \text{Pic}(X).$$

In particular, $\text{Cl}(\mathbb{CP}^1) \cong \text{Pic}(\mathbb{CP}^1)$.

Proof. This is a standard theorem; see [4, Chapter II, Proposition 6.15] for the algebraic-geometric formulation on smooth curves. $\square$

9.4 Computation of the Picard group of $\mathbb{CP}^1$

From Theorem 8.4 we already know that

$$\mathrm{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

Combining this with Theorem 9.7, we immediately obtain the main result of this chapter.

Theorem 9.8. *The Picard group of the complex projective line is isomorphic to the additive group of integers: $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$.*

Proof. By Theorem 8.4,

$$\mathrm{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}.$$

By Theorem 9.7, $\mathrm{Cl}(\mathbb{CP}^1) \cong \mathrm{Pic}(\mathbb{CP}^1)$. Therefore $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$. $\square$

Line bundles on $\mathbb{CP}^1$ are classified by a single integer. Just as divisor classes on $\mathbb{CP}^1$ are determined by degree, line bundles on $\mathbb{CP}^1$ are determined by degree as well. Equivalently, every line bundle on $\mathbb{CP}^1$ is isomorphic to one of the form $\mathcal{O}(n)$ for a unique integer n .

10 Summary

We began with the classical problem of understanding roots of complex polynomials and developed it into a global geometric theory on the Riemann sphere. Then, passing from $\mathbb{C}$ to $\mathbb{CP}^1$, we associate to each nonzero meromorphic function a divisor.

From this construction, two exact sequences arose naturally: one describing principal divisors as rational functions modulo nonzero constants, and one describing divisor classes as divisors modulo principal divisors. The central theorem, Theorem 7.3, proved that on $\mathbb{CP}^1$, $\mathrm{Prin}(\mathbb{CP}^1) = \ker(\deg)$. This made it possible to compute the divisor class group and obtain $\mathrm{Cl}(\mathbb{CP}^1) \cong \mathbb{Z}$. Using the relation between divisor classes and line bundles, this further led to the identification $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$.

Recall that on general smooth curves, divisors of degree zero need not be principal, and the Picard group becomes much richer. In this sense, the complex projective line is the first and most transparent case of a broader theory of divisors, line bundles, and algebraic curves.

Future Directions

The results obtained in this paper suggest several natural directions for further study. The first is to move beyond the special case of $\mathbb{CP}^1$ and consider arbitrary smooth curves, where the equality $\mathrm{Prin}(X) = \ker(\deg)$ no longer holds in general. Divisors of degree zero need not be principal, and the divisor class group becomes a richer invariant.

A second direction is the study of the Picard group on general smooth curves. While on $\mathbb{CP}^1$ one has the isomorphism $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$, for higher genus curves the Picard group leads naturally to the study of Jacobians and moduli of line bundles.

Finally, the constructions in this paper also point toward a more sheaf-theoretic and cohomological perspective. In more advanced treatments, divisor classes and line bundles are related through the theory of invertible sheaves, and the Picard group may be interpreted in terms of cohomological invariants.

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